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PMRNet: Physics-informed Multi-scale Refinement Network for Medical Image Segmentation

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Abstract

*Medical image segmentation demands both high accuracy and computational efficiency, yet existing methods face a critical trade-off: CNNs lack global context while transformers incur prohibitive costs for deployment on resource-constrained devices. To address this challenge, we propose a **P**hysics-informed **M**ulti-scale **R**efinement Network (PMRNet), integrating symplectic geometry, renormalization group theory, and entropy diffusion to guide feature learning. PMRNet features three innovations: (1) a physics-informed encoder with Enhanced Symplectic Convolution for boundary detection and Renormalization Group-inspired Downsampling for information preservation; (2) a Pseudo-Global Receptive Field module achieving near-global context with linear complexity through entropy-driven diffusion; and (3) a boundary-aware decoder for precise delineation. With only 0.87M parameters and 3.43 GFLOPs, PMRNet achieves 87.25% IoU and 92.56% Dice on the challenging Clinic dataset, outperforming state-of-the-art (SOTA) models with even 100× more parameters across 12 medical imaging datasets while maintaining computational efficiency. Code is available at* https://github.com/KangBoce/PMRNet.

1. Introduction

Medical image segmentation supports a range of clinical tasks, from automated diagnosis to treatment planning and population-level screening. Models deployed at the point of care or on portable hardware must be accurate without being computationally expensive, and most existing architectures fall short on one side or the other.

UNet [51] defined the encoder-decoder blueprint that much of the field still builds on. Its variants [31, 34, 38, 45, 73, 74] improved feature reuse and attention gating, but the underlying architectures remain heavy enough that running them on embedded devices is impractical. Vision transformers [8, 9, 17, 24, 25, 72] later showed that self-attention could capture spatial relationships well. However, this expressiveness comes at significant computational cost: for example, TransUNet requires 105M+ parameters, making deployment on resource-constrained devices impractical.

On the efficiency side, UNeXt [61] and MALUNet [52] demonstrated that models under 2M parameters can produce reasonable segmentations. However, their segmentation quality degrades noticeably on small structures, limiting clinical utility. More recent designs like EMCAD [50] and Deformable-LKA [3] fold in specialized attention or large kernels to recover some of that lost accuracy, but the improvements are uneven across imaging modalities, and the extra complexity is likely to slow inference speed.

Among methods that do achieve high accuracy, the usual response to weak inductive bias has been to add more parameters. Instead of scaling up, we take an opposite route. Inspired by physics principles, we propose PMRNet (**P**hysics-informed **M**ulti-scale **R**efinement Network), a lightweight architecture whose main contributions are:

Physics-informed Encoder: A hierarchical encoder built around Enhanced Symplectic Convolution (ESConv). By treating feature maps as position-momentum pairs, ESConv stabilizes gradient flow and sharpens boundary responses. The encoder also includes Progressive Refinement modules, Adaptive Scale Selectors with rotation-augmented convolution, Lightweight Memory Banks for cross-scale prototype matching [56], and a downsampling scheme inspired by renormalization group to preserve multi-scale information.

Efficient Global Context Modeling: Pseudo-Global Receptive Field (PGRF) models feature propagation through the heat equation [47], reaching near-global context in $O(N)$ time without self-attention.

A boundary-aware decoder where fusion modules gate skip connections based on the local boundary strength, keeping the decoder spatially precise and with few parameters.

PMRNet has 0.87M parameters and costs 3.43 GFLOPs in total. Across 12 benchmarks covering ultrasound, endoscopy, dermoscopy, and histopathology, it outperforms models that are 10 to 100 times larger, while maintaining real-time inference speeds tested in Section 4.3.

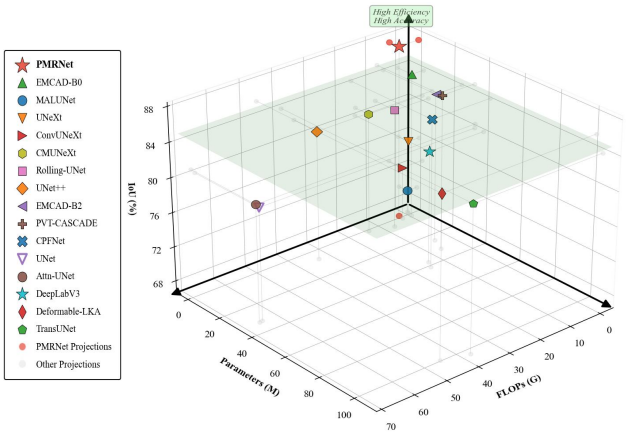


Figure 1. Computational Complexity Comparison: IoU vs. GFlops and Parameters on the Clinic [4] dataset. The green plane at 85% IoU marks the high-accuracy threshold.

2. Related Work

2.1. Boundary Preservation and Information Loss

Segmenting boundaries accurately is difficult when objects are small or in irregular shapes. UNet's skip connections [51] help by forwarding features to the decoder, and later extensions like UNet++ [74] added nested paths to sharpen these connections. The limitation is that standard convolutions respond weakly to sharp intensity gradients that define boundaries. Classical operators [7, 36, 66] detect edges reliably in isolation, but perform poorly inside learned pipelines because they compute fixed gradients without reference to the features being processed. Learned edge attention [5, 59] partially addresses this through making boundary detection task-adaptive, yet it still treats edge signals as an auxiliary input rather than coupling them with the feature update itself. This leaves a gap that boundary information and semantic features are processed in separate streams.

We address this gap from two angles. For feature extraction, we design a module that pairs spatial features with their derivatives as a coupled system inspired by symplectic geometry [2, 71]. For downsampling, we draw on the coarse-graining principle of renormalization group theory [65], folding fine-scale structure into effective coarser representations rather than discarding it.

2.2. Efficient Global Context Modeling

Understanding what a local patch represents often requires information from distant regions. Transformers [8, 9, 17, 24, 25, 72] solve this with self-attention, but the $O(N^2)$ cost limits practical deployment. Large-kernel [3, 16, 41] and dilated convolutions [68] widen the receptive field more cheaply, though coverage remains partial. Multi-scale aggregation [39, 50, 70] brings incremental gains that tend to plateau.

Diffusion processes are an alternative the segmentation community has largely overlooked. The heat equation propagates information at linear cost, and spatially varying coefficients naturally preserve boundaries [47]. Our PGRF module builds on this principle, spreading context through a learned adaptive diffusion process in the bottleneck without any attention mechanism.

2.3. Lightweight and Efficient Design

Models intended for resource-constrained settings, such as embedded devices or hospital workstations, must balance accuracy against strict computational budgets. UNeXt [61] showed MLP-based token mixing could replace heavy convolutional blocks, and CMUNeXt [60] and ConvUNeXt [23] continued in this direction. These models are genuinely small, but the accuracy gap on challenging cases is hard to ignore. At the other extreme, SAM [37] achieves strong zero-shot performance despite requiring 632M+ parameters, which is limited for clinical practice. Although SAM-based methods are not the central focus of this work, they illustrate how far current accuracy depends on scale. Methods like EMCAD [50] and Rolling-UNet [40] sit between these extremes, adding modules that help on some modalities while slowing inference on others.

PMRNet follows a different strategy, relying on physics-informed priors to maintain competitive performance at only 0.87M parameters and 3.43 GFLOPs.

3. Methodology

In this section, we begin by introducing a basic building block, followed by the encoder architecture, bottleneck design, and decoder structure. The overall architecture is illustrated in Figure 2.

3.1. Efficient Convolution Block

We first introduce the Efficient Convolution Block (ECB), a fundamental building block used throughout our architecture. ECB provides lightweight feature refinement through a streamlined design consisting of a depthwise separable convolution [12, 29, 53] with 3×3 kernel, followed by batch normalization [33] and SiLU activation [18]:

$$\mathrm{ECB}(\mathbf{x}) = \mathrm{SiLU}(\mathrm{BN}(\mathrm{DSConv}_{3\times3}(\mathbf{x}))) \quad (1)$$

ECB serves as a basic building component in multiple network modules including the Progressive Refinement Unit (PR), Boundary-aware Fusion (BF), and as a standalone refinement block in the decoder stages, enabling uniform feature processing with minimal computational overhead.

3.2. Encoder Architecture

The encoder consists of three stages with progressively increasing abstraction. Each stage integrates four key com-

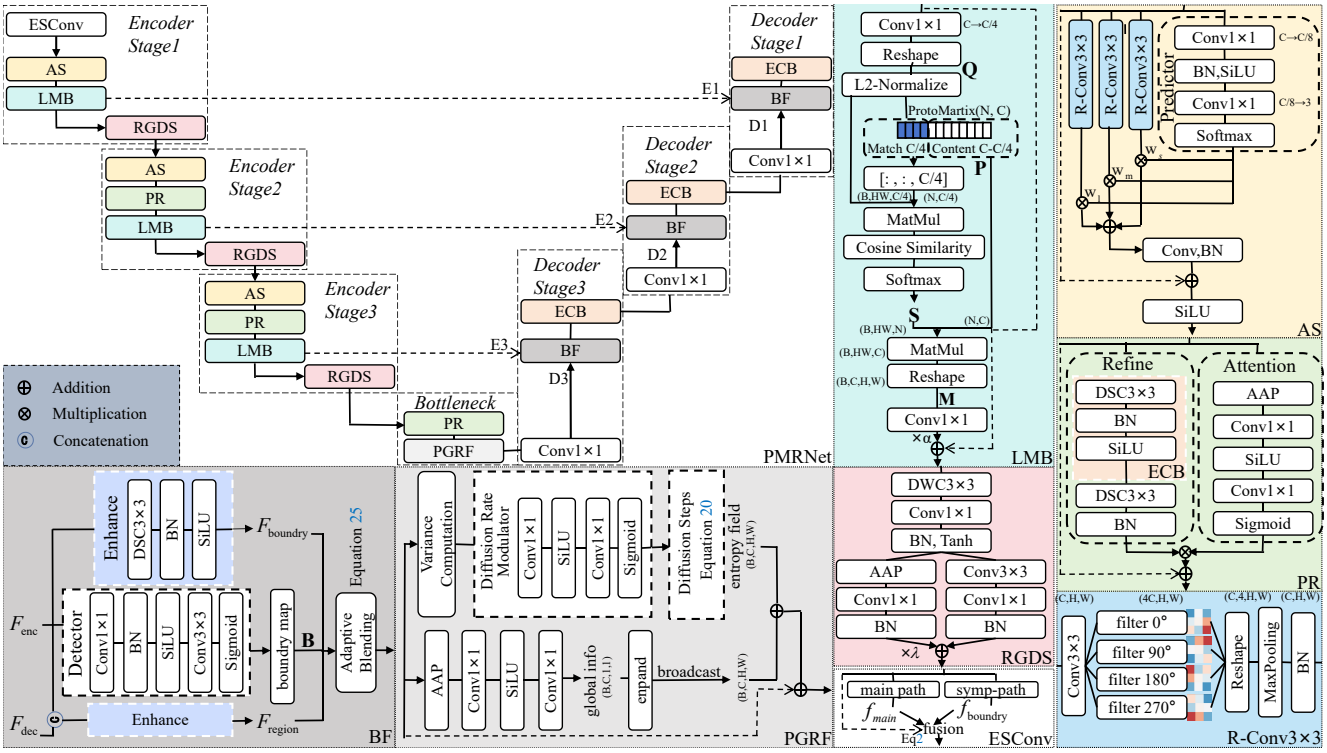


Figure 2. Overview of the proposed architecture: Enhanced Symplectic Convolution (ESConv), Progressive Refinement Unit (PR), Adaptive Scale Selector (AS), Lightweight Memory Bank (LMB), Renormalization Group-inspired Downsampling (RGDS), Pseudo-Global Receptive Field (PGRF), Boundary-aware Fusion (BF), Efficient Convolution Block (ECB).

ponents: ESConv (stage 1) or PR module (stages 2-3) for feature processing, AS module for multi-angle feature extraction, LMB module for hierarchical prototype-based retrieval [56], and RGDS module for scale transformation.

3.2.1. Enhanced Symplectic Convolution

Boundary pixels in medical images are often only a few pixels wide, making them easy to miss by standard convolutions that treat interior and edge regions identically. A natural fix is to inject explicit gradient information: Sobel [36] or Canny [7] filters can localize intensity transitions, and learned edge attention [5, 64] can adapt to the task. When we replaced ESConv with these alternatives in Section 5.3, all three degraded both IoU and HD95 on ISIC2017 [14]. The fixed-operator variants suffered most, suggesting that rigid gradient filters conflict with learned features at different network stages. Even learned edge attention, though closer to ESConv than the fixed operators, still fell short by 0.93% IoU, indicating that boundary detection alone, without coupling to the feature update, leaves accuracy.

These results motivated a design that keeps gradient information coupled with spatial features throughout the update, rather than injecting it as a separate signal. We take a cue from Hamiltonian mechanics, where position and momentum are updated jointly to prevent either from being lost [2, 22]. ESConv pairs the feature map (position) with its spatial gradient (momentum) and applies a coupled residual step whose structure resembles a first-order symplectic integrator. The coupling is approximate, since batch normalization and $\tanh$ break strict symplectic structure, but it serves the practical goal of preventing gradient signals from being washed out during training.

Given input features $\mathbf{x} \in \mathbb{R}^{C\times H\times W}$, ESConv employs a dual-path architecture:

$$\text{ESConv}(\mathbf{x}) = \text{Fusion}(\mathbf{f}_{\text{main}}, \mathbf{f}_{\text{boundary}}) + k \cdot \mathbf{x} \quad (2)$$

where $\mathbf{f}_{\text{main}} = \text{GELU}(\text{BN}(\text{DSConv}_{3\times3}(\mathbf{x})))$ captures semantic features through depthwise separable convolution with GELU activation [28], and k is a learnable residual weight [26].

The boundary enhancement path begins by computing spatial gradients in both directions using Sobel-initialized grouped convolutions [36]. The gradient magnitude is approximated with the L1 norm for efficiency, $\|\nabla\mathbf{x}\| \approx (|\mathbf{G}_x * \mathbf{x}| + |\mathbf{G}_y * \mathbf{x}|)/\sqrt{2}$, and then refined through a lightweight block to produce enhanced gradient features $\hat{\nabla}\mathbf{x} = \text{GELU}(\text{BN}(\text{DSConv}_{3\times3}(\|\nabla\mathbf{x}\|)))$. These are concatenated with the original features to form the augmented

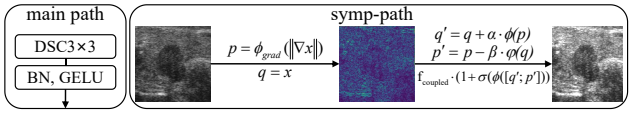


Figure 3. Architecture and feature visualization of the ESConv module. The symplectic path derives boundary cues from spatial gradients (middle) and yields enhanced edge responses after the coupled update (right).

representation:

$$\mathbf{x}_{\text{aug}} = [\mathbf{x};\, \hat{\nabla}\mathbf{x}] \in \mathbb{R}^{2C\times H\times W} \tag{3}$$

Channel splitting decomposes $\mathbf{x}_{\text{aug}}$ into $\mathbf{q}, \mathbf{p} = \text{Split}(\mathbf{x}_{\text{aug}})$, where $\mathbf{q} \in \mathbb{R}^{C\times H\times W}$ retains the original spatial features and $\mathbf{p} \in \mathbb{R}^{C\times H\times W}$ encodes the gradient-derived boundary information. Following the Hamiltonian analogy, the coupled update reorders the pair as $[\mathbf{p}; -\mathbf{q}]$ and applies a joint transformation:

$$\begin{bmatrix}\mathbf{q}'\\ \mathbf{p}'\end{bmatrix} = \begin{bmatrix}\mathbf{q}\\ \mathbf{p}\end{bmatrix} + \epsilon \cdot \tanh\left(\text{BN}(\text{Conv}_{1\times1}([\mathbf{p}; -\mathbf{q}]))\right) \tag{4}$$

where $\epsilon = 0.1$ controls the step size of the coupled update. A lightweight detector then predicts a boundary map $\mathbf{D} = \sigma(\phi_{\text{boundary}}([\mathbf{q}'; \mathbf{p}'])) \in \mathbb{R}^{1\times H\times W}$ through convolutional layers with channel reduction to $C/4$, which modulates the coupled features:

$$\mathbf{f}_{\text{boundary}} = \mathbf{f}_{\text{coupled}} \odot (1 + \mathbf{D}) \tag{5}$$

The two paths are fused through concatenation and 1×1 convolution projection to produce the final output.

3.2.2. Progressive Refinement Unit

The PR module is employed in encoder stages 2 and 3, as well as in the bottleneck, to perform iterative feature enhancement. Given input features, PR executes T iterations ($T = 2$ for stage 2, $T = 3$ for stage 3 and bottleneck) of gated residual refinement. Each iteration t applies one ECB module followed by an additional batch normalization [33] and depthwise separable convolution [29]:

$$\mathbf{r}^{(t)} = \text{BN}(\text{DSConv}_{3\times3}(\text{ECB}(\mathbf{x}^{(t-1)}))) \tag{6}$$

Followed by adaptive gating through global average pooling, two 1×1 convolutions with channel reduction (to $C/16$), and sigmoid activation to produce attention weights $\mathbf{a}^{(t)} \in [0, 1]^C$ inspired by squeeze-and-excitation [30]. The residual update then combines the refined features with the gated residual:

$$\mathbf{x}^{(t)} = \mathbf{x}^{(t-1)} + \mathbf{r}^{(t)} \odot \mathbf{a}^{(t)} \tag{7}$$

We set T by validation and we find that increasing beyond 3 iterations gave no further gain on PH^2 [44] dataset.

3.2.3. Adaptive Scale Selector with Rolling-Convolution

Small objects in medical images often exhibit arbitrary orientations, requiring robust multi-angle feature extraction. Traditional data augmentation with rotations [54] increases training time without providing consistent orientation handling. Proposed AS module addresses this through efficient rotation-augmented convolution that processes features at multiple orientations simultaneously. Given an input feature map $\mathbf{x} \in \mathbb{R}^{C\times H\times W}$, instead of explicitly rotating features four times requiring 12 rotation operations per forward pass, we employ grouped convolution with $4C$ output channels where the weight tensor $\mathbf{W}_{\text{rot}} \in \mathbb{R}^{4C\times C\times3\times3}$ is initialized such that for each input channel i:

$$\mathbf{W}_{\text{rot}}[4i + j, i, :, :] = \text{Rot90}(\mathbf{W}^i_{\text{base}}, j) \text{ for } j \in \{0, 1, 2, 3\} \tag{8}$$

The result is $4\times$ parameter sharing across rotation angles while reducing computational cost from $O(4CHW)$ to $O(CHW)$. The AS module employs three parallel rotation-augmented branches capturing small, medium, and large receptive fields. Although these branches target different scales, for computational efficiency they all use the same 3×3 Rolling-Convolution (R-Conv), with scale differentiation achieved through learned weighting. The multi-scale fusion can be expressed as:

$$\mathbf{f}_{\text{multi}} = \sum_{s\in\{\text{small, medium, large}\}} \text{R-Conv}^{(3\times3)}(\mathbf{x}) \cdot \alpha_s(\mathbf{x}) \tag{9}$$

Then, they are fused with the residual connection [26]:

$$\text{AS}(\mathbf{x}) = \text{SiLU}\left(\text{Fusion}(\mathbf{f}_{\text{multi}}) + \mathbf{x}\right) \tag{10}$$

where $\alpha_s(\mathbf{x}) = \text{Softmax}(\text{Predictor}(\mathbf{x}))$ are learnable scale weights. These adaptive weights enable dynamic balancing of contributions from different scale representations, allowing the network to emphasize the most relevant scale for each spatial location.

3.2.4. Lightweight Memory Bank

To enable cross-scale feature matching across encoding stages, we design LMB module, which stores N learnable prototypes $\mathbf{P} \in \mathbb{R}^{N\times C}$ at each encoder stage ($N = 8, 12, 16$ for stages 1–3 respectively), inspired by prototypical networks [27, 56, 63]. More prototypes are allocated to deeper stages to match their greater semantic variety. Given input $\mathbf{x}$, we compute compact query representations through 1×1 convolution with channel reduction to $C/4$ and perform temperature-scaled cosine similarity matching with stored prototypes. The retrieved prototype features are then folded back into the input through a learnable residual connection:

$$\mathbf{Q} = \text{Reshape}(\text{Conv}_{1\times1}(\mathbf{x})) \in \mathbb{R}^{HW\times C/4} \tag{11}$$

$$S = \text{Softmax}\left(10 \cdot \frac{\mathbf{Q}}{\|\mathbf{Q}\|} \cdot \frac{\mathbf{P}^T_{:,:C/4}}{\|\mathbf{P}_{:,:C/4}\|}\right) \in \mathbb{R}^{HW \times N} \quad (12)$$

$$\mathbf{M} = \text{Reshape}(S \cdot \mathbf{P}) \in \mathbb{R}^{C \times H \times W} \quad (13)$$

$$\text{LMB}(\mathbf{x}) = \mathbf{x} + \alpha \cdot \text{Conv}_{1\times1}(\mathbf{M}) \quad (14)$$

where $\|\cdot\|$ denotes the L_2 norm, the temperature scaling factor 10 sharpens attention weights for focused prototype retrieval, and α is a learnable residual coefficient initialized at 0.1 to ensure stable training.

3.2.5. Renormalization Group-Inspired Downsampling

Max-pooling and strided convolutions reduce resolution by selecting or averaging local patches, treating all spatial positions equally. For small objects that occupy only a handful of pixels, this uniform reduction can erase the very features the decoder needs to recover.

The renormalization group (RG) in statistical physics [65] faces an analogous problem when coarse-graining a lattice: naively averaging neighboring sites discards short-range correlations that affect the system's large-scale behavior. The standard remedy is to absorb those correlations into renormalized coupling constants before reducing the lattice spacing, so that the coarser description remains faithful to the original. RGDS borrows this idea. Before halving the spatial dimensions, a learned coupling function modifies the feature map so that fine-scale structure is encoded into the channel activations rather than lost. The coupling function plays a role analogous to the renormalized constants in RG: it lets the network decide, per channel, what local structure is worth keeping before resolution is halved:

$$\mathbf{x}_{\text{enhanced}} = \mathbf{x} + \alpha \cdot \big(\boldsymbol{\beta}(\mathbf{x}) + \boldsymbol{\gamma}(\mathbf{x}) \odot \mathbf{x}\big) \quad (15)$$

where $\boldsymbol{\beta}$ and $\boldsymbol{\gamma}$ provide additive and multiplicative corrections respectively, produced jointly by a depthwise 3×3 convolution, 1×1 projection, batch normalization [33], and $\tanh$ activation. The step size α is learnable and initialized to 0.1.

The enriched features are then downsampled through two complementary paths:

$$\mathbf{f}_{\text{low}} = \text{SiLU}(\text{BN}(\text{Conv}^{\text{stride}=2}_{3\times3}(\mathbf{x}_{\text{enhanced}}))) \quad (16)$$

$$\mathbf{f}_{\text{high}} = \text{BN}(\text{Conv}_{1\times1}(\text{AvgPool}_{2\times2}(\mathbf{x}_{\text{enhanced}}))) \quad (17)$$

$$\text{RGDS}(\mathbf{x}) = \mathbf{f}_{\text{low}} + \lambda \cdot \mathbf{f}_{\text{high}} \quad (18)$$

where the strided convolution path captures low-frequency semantics, the average pooling path retains higher-frequency detail, and $\lambda = 0.3$ balances the two.

3.3. Bottleneck Architecture

The bottleneck layer integrates two critical components for global context modeling. The Progressive Refinement (PR) module performs $T = 3$ iterations as described previously, followed by the Pseudo-Global Receptive Field (PGRF) module. Global context is essential for disambiguating small objects from background clutter, but self-attention mechanisms [62] incur prohibitive quadratic complexity. We draw inspiration from entropy-driven diffusion in physical systems [47], where information naturally flows from high-certainty regions to uncertain areas. This principle is appropriate for object detection: confident features should propagate to ambiguous regions while respecting object boundaries. Our proposed PGRF module implements this through adaptive diffusion governed by the heat equation with learned coefficients:

$$\frac{\partial \mathbf{u}}{\partial t} = D(\mathbf{x}) \cdot \nabla^2 \mathbf{u} + \mathbf{g}(\mathbf{x}) \quad (19)$$

where $\mathbf{u}$ is the evolving feature map, $D(\mathbf{x})$ is a data-adaptive diffusion rate learned from local variance, $\nabla^2 \mathbf{u}$ represents the Laplacian operator, and $\mathbf{g}(\mathbf{x})$ provides global context injection. We discretize this partial differential equation using explicit Euler integration with 3 steps ($k \in \{0, 1, 2\}$):

$$\mathbf{u}^{(k+1)} = \mathbf{u}^{(k)} + D(\mathbf{x}) \odot \mathcal{L}(\mathbf{u}^{(k)}) \cdot (0.5)^k + 0.1 \cdot \mathbf{g} \quad (20)$$

where the Laplacian is efficiently approximated through depthwise separable convolution [29] as $\mathcal{L}(\mathbf{u}) = \text{DSConv}_{3\times3}(\mathbf{u}) - \mathbf{u}$ and $(0.5)^k$ provides exponential decay to prevent over-smoothing. The adaptive diffusion coefficient is computed from local variance through average pooling operations:

$$D(\mathbf{x}) = \sigma(\text{Conv}_{1\times1}(\text{SiLU}(\text{Conv}_{1\times1}(\text{Var}(\mathbf{x}))))) \quad (21)$$

where variance is estimated as $\text{Var}(\mathbf{x}) = \text{AvgPool}_{3\times3}((\mathbf{x} - \text{AvgPool}_{3\times3}(\mathbf{x}))^2)$, assigning high diffusion rates to homogeneous regions and low rates near boundaries. This enables information propagation from confident to uncertain regions. Global features $\mathbf{g}$ are extracted via adaptive average pooling to 1×1, followed by two 1×1 convolutions with channel reduction to $C/4$ and SiLU activation, then broadcast to spatial dimensions, providing consistent contextual guidance throughout the diffusion process.

3.4. Decoder Architecture

Each decoding stage progressively upsamples features through three stages, each using 1×1 convolution for channel reduction before fusion. The decoder employs Boundary-aware Fusion (BF) for edge-sensitive feature aggregation between decoder and encoder pathways, followed by an

Table 1. Comprehensive Performance on TG3K and Clinic Datasets. All metrics are reported as percentages (%) except HD95 and ASD (in pixels). ↑ indicates higher is better, ↓ indicates lower is better. Best results are in **bold**.

Method	Params(M)	FLOPs(G)	TG3K								Clinic							
			IoU↑	Dice↑	HD95↓	ASD↓	SE↑	PC↑	SP↑	ACC↑	IoU↑	Dice↑	HD95↓	ASD↓	SE↑	PC↑	SP↑	ACC↑
UNet [51]	34.53	65.53	71.20	80.90	23.19	7.81	83.81	83.63	98.83	97.43	80.00	87.26	21.61	6.63	87.33	91.16	99.11	98.21
UNet++ [74]	9.16	34.65	72.47	81.55	20.14	7.58	83.94	85.07	98.94	97.57	82.05	88.84	23.85	5.86	90.74	89.41	99.01	98.33
TransUNet [9]	105.32	38.52	73.22	82.24	20.97	7.07	84.69	84.94	98.72	97.46	83.87	90.35	16.72	4.27	91.14	91.48	99.25	98.50
UNeXt [61]	1.47	0.57	70.64	80.06	19.19	7.19	84.44	82.16	98.56	97.45	75.75	84.40	29.48	6.84	85.99	87.09	98.94	97.92
Rolling-UNet [40]	7.10	8.28	72.09	81.20	18.10	6.71	83.84	83.69	98.81	97.49	80.98	87.82	19.55	5.63	88.75	89.63	99.15	98.43
PVT-CASCADE [49]	34.12	7.62	73.77	82.75	17.34	5.93	86.32	83.97	98.67	97.44	85.09	91.07	14.94	4.38	91.10	92.03	99.34	98.77
Attn-UNet [45]	34.88	66.64	73.29	82.11	19.01	7.40	83.66	86.20	**99.01**	97.56	80.52	87.46	23.03	7.17	88.43	90.21	99.06	98.18
Deformable-LKA [3]	101.64	46.12	73.04	82.25	17.94	5.99	85.30	83.30	98.64	97.60	85.65	91.60	13.90	3.57	92.19	92.15	99.18	98.70
DeepLabV3 [10]	39.76	14.92	66.21	76.92	19.93	7.76	79.09	81.30	98.74	97.12	80.09	87.50	19.48	6.47	89.17	87.63	99.00	98.30
CPFNet [19]	43.27	16.15	73.89	83.02	16.10	5.67	85.43	84.90	98.76	97.60	84.26	90.70	16.24	4.02	91.72	91.26	99.16	98.58
EMCAD-B2 [50]	26.76	5.60	74.88	83.56	15.49	5.75	**86.35**	84.59	98.75	97.83	84.30	90.47	14.47	4.31	90.61	92.35	99.35	98.71
CMUNeXt [60]	3.15	14.84	69.48	79.13	19.91	8.05	82.28	83.20	98.61	97.17	80.97	87.92	21.16	6.72	89.07	89.95	98.90	98.19
EMCAD-B0 [50]	3.92	0.84	70.83	80.43	20.01	7.08	82.87	84.34	98.75	97.18	83.98	90.52	16.35	4.45	90.39	91.80	99.36	98.68
MALUNet [52]	0.18	0.17	66.67	77.00	20.63	7.66	81.18	79.11	98.24	96.78	69.59	79.69	29.20	8.94	82.52	81.18	98.52	97.18
ConvUNeXt [23]	3.54	3.69	64.76	74.93	22.72	8.76	78.15	79.62	98.58	96.82	73.20	81.50	26.12	9.10	82.14	85.07	98.82	97.78
PMRNet **(Ours)**	0.87	3.43	**76.19**	**84.73**	**14.96**	**5.51**	86.03	**87.61**	98.97	**97.85**	**87.25**	**92.56**	**13.13**	**3.54**	**93.40**	**92.50**	**99.38**	**98.98**

Table 2. Performance Comparison on other 10 Medical Image Datasets. All metrics are reported as percentages (%) except HD95 (in pixels). ↑ indicates higher is better, ↓ indicates lower is better. For brevity, we only report top-performing methods with 3 key metrics: IoU, Dice, and HD95. Best results are in **bold**.

Method	Breast			BUSI			DDTI			DSB18			GlaS		
	IoU↑	Dice↑	HD95↓	IoU↑	Dice↑	HD95↓	IoU↑	Dice↑	HD95↓	IoU↑	Dice↑	HD95↓	IoU↑	Dice↑	HD95↓
UNet	60.53	71.55	40.35	64.74	74.24	36.76	80.18	88.64	20.48	85.83	92.07	3.68	82.73	90.13	25.91
UNet++	61.70	72.48	38.11	67.89	76.91	35.52	80.30	88.75	19.21	85.15	91.59	3.98	84.79	91.46	21.18
TransUNet	62.64	72.67	29.52	69.06	78.22	28.67	80.80	89.11	19.63	86.25	92.26	3.40	83.98	90.83	24.37
Rolling-UNet	63.64	73.65	31.82	71.55	80.78	26.64	79.06	87.94	20.34	84.84	91.42	4.04	84.59	91.29	23.25
EMCAD-B2	64.72	74.82	30.38	72.03	80.61	22.69	82.27	90.02	16.98	85.33	91.71	3.83	85.39	91.79	21.44
Deformable-LKA	38.27	51.09	67.32	71.83	80.63	23.33	**82.81**	90.30	16.07	85.86	92.00	3.94	83.84	90.94	25.51
DeepLabV3	56.98	68.69	32.48	71.56	80.60	23.34	82.80	**90.32**	**15.90**	66.57	78.66	8.56	79.84	88.37	21.84
PMRNet **(Ours)**	**64.73**	**75.51**	**26.03**	**73.56**	**81.88**	**22.58**	82.52	90.17	16.70	**86.87**	**92.74**	**3.36**	**85.70**	**92.03**	**19.58**

Method	ISIC2017			ISIC2018			Kvasir-SEG			PH^2			TN3K		
	IoU↑	Dice↑	HD95↓	IoU↑	Dice↑	HD95↓	IoU↑	Dice↑	HD95↓	IoU↑	Dice↑	HD95↓	IoU↑	Dice↑	HD95↓
UNet	81.68	88.57	16.70	80.73	88.03	20.23	71.68	80.37	45.10	88.07	93.47	18.52	68.68	78.63	30.37
UNet++	81.05	87.98	16.81	81.65	88.81	18.41	76.41	84.02	37.76	89.70	94.46	14.82	70.13	79.71	29.99
TransUNet	82.06	88.85	15.70	80.68	87.90	19.01	78.36	85.64	30.98	89.69	94.39	18.86	70.63	80.20	27.55
Rolling-UNet	82.14	88.82	14.12	81.67	88.76	17.70	76.11	83.88	32.55	89.08	94.08	17.86	70.13	79.77	27.81
EMCAD-B2	81.49	88.29	15.27	81.78	88.74	16.58	78.62	85.85	29.71	89.92	94.58	14.68	72.72	82.02	24.34
Deformable-LKA	78.76	86.02	18.98	82.47	89.26	**15.78**	77.31	84.37	31.32	90.45	94.90	14.60	67.04	77.69	32.16
DeepLabV3	81.28	88.44	15.56	81.14	88.43	18.04	73.62	81.92	33.04	88.86	93.99	15.05	67.01	77.62	27.59
PMRNet **(Ours)**	**83.14**	**89.69**	**13.73**	**82.51**	**89.32**	15.79	**81.21**	**87.67**	**26.68**	**91.07**	**95.24**	**14.13**	**73.50**	**82.50**	**23.03**

ECB module for lightweight feature refinement. The BF adaptively fuses decoder features $\mathbf{f}_{\text{dec}}$ and encoder features $\mathbf{f}_{\text{enc}}$ according to learned boundary strength. We first extract a boundary map:

$$\mathbf{B} = \sigma(\text{Conv}_{3\times3}(\text{SiLU}(\text{BN}(\text{Conv}_{1\times1}(\mathbf{f}_{\text{enc}}))))) \quad (22)$$

It identifies edge locations through a lightweight detector with channel reduction to $C/8$. Two parallel processing paths then handle region and boundary features separately using an enhancement module (Enhance) consisting of $\text{DSConv}_{3\times3}$-BN-SiLU:

$$\mathbf{f}_{\text{region}} = \text{Enhance}([\mathbf{f}_{\text{dec}}^{\uparrow}; \mathbf{f}_{\text{enc}}]) \quad (23)$$

where $\mathbf{f}_{\text{dec}}^{\uparrow}$ denotes bilinearly upsampled decoder features.

$$\mathbf{f}_{\text{boundary}} = \text{Enhance}(\mathbf{f}_{\text{enc}}) \quad (24)$$

The final output dynamically balances these paths based on spatial boundary strength:

$$\text{BF}(\mathbf{f}_{\text{dec}}, \mathbf{f}_{\text{enc}}) = \mathbf{f}_{\text{region}} \odot (1 - \mathbf{B}) + \mathbf{f}_{\text{boundary}} \odot \mathbf{B} \quad (25)$$

The final segmentation output is produced through an additional ECB followed by dropout and a 1×1 convolution that projects to the number of classes.

4. Experiments

4.1. Datasets and Evaluation Metrics

We utilized 12 publicly available medical image segmentation datasets that have been benchmarked in recent studies. The datasets include Breast [67], BUSI [1], Clinic [4], DDTI [46], DSB2018 [6], GlaS [55], ISIC2017 [14], ISIC2018 [13], Kvasir-SEG [35], PH^2 [44], TN3K [20], TG3K [11]. These datasets encompass various medical imaging modalities including ultrasound, endoscopy, dermoscopy, and histopathology images, captured under different conditions and featuring diverse anatomical structures and pathologies.

To assess performance, we utilized a broad range of metrics. For pixel-based evaluation, we used Intersection over Union (IoU) [69], Dice coefficient [15], Sensitivity (SE), Precision (PC), Specificity (SP), and Accuracy (ACC) [48]. For surface distance-based metrics, the 95th percentile of the Hausdorff distance (HD95) [32] and the Average Surface Distance (ASD) [58] were used to provide a comprehensive evaluation of boundary delineation quality.

4.2. Experimental Setup and Implementation Details

All experiments were conducted on a single NVIDIA RTX 4090 GPU. To ensure a fair comparison, every model was trained for 200 epochs with a batch size of 8. We employed the AdamW optimizer [42] with an initial learning rate of 0.001, which was dynamically adjusted using a cosine annealing schedule. The loss function was a balanced combination of Binary Cross-Entropy (BCE) [21] and Dice loss [57]:

$$\mathcal{L} = 0.5 \cdot \mathcal{L}_{\text{BCE}} + 0.5 \cdot \mathcal{L}_{\text{Dice}} \tag{26}$$

We used identical hardware and software configurations. All reported metrics represent the median values from three independent training runs, and the best model checkpoint was selected based on validation set performance.

4.3. Results and Comparison

Tables 1 and 2 summarize segmentation performance across all 12 datasets. PMRNet consistently achieves competitive or superior performance compared to 15 baseline methods, including: (1) classical encoder-decoder architectures and their variants (UNet [51], UNet++ [74], DeepLabV3 [10]); (2) attention-based methods (TransUNet [9], Attn-UNet [45], CPFNet [19]); (3) lightweight architectures (UNeXt [61], EMCAD-B0 [50], MALUNet [52], ConvUNeXt [23], CMUNeXt [60]); and (4) recent state-of-the-art models (PVT-CASCADE [49], EMCAD-B2 [50], Rolling-UNet [40], Deformable-LKA [3]).

Our method demonstrates quite strong performance on challenging datasets with small objects. For example, on the Clinic dataset [4], PMRNet achieves 87.25% IoU and 92.56% Dice score, outperforming the second-best method by significant margins. Similarly, on TG3K [11] and Kvasir [35] datasets, PMRNet improves over the second-best method on both region-based metrics (IoU, Dice) and boundary-based metrics (HD95, ASD), validating the effectiveness of our physics-informed design.

Additionally, we evaluated the inference speed of recent SOTA methods (post-2021) on the ISIC2017 dataset using a single NVIDIA RTX 4090 GPU. As shown in the FPS comparison, PMRNet achieves 152.03 FPS, outperforming EMCAD (146.41 FPS), TransUNet (96.37 FPS), PVT-CASCADE (104.59 FPS), Rolling-UNet (33.90 FPS), and Deformable-LKA (24.73 FPS), PMRNet is both the fastest and the most accurate among these recent methods.

Figure 4 presents qualitative comparisons with baseline methods. PMRNet produces more precise boundary delineation, especially for small objects with complex shapes.

5. Ablation Study

5.1. Component Ablation Study

To clarify the contribution of each key component in PMRNet, we firstly conduct ablation experiments by removing individual modules while keeping all other components unchanged. The results on PH^2 [44] dataset are reported in Table 3.

Table 3. Component ablation study on PH^2 [44] dataset.

Model Variant	IoU(%)↑	Dice(%)↑	HD95↓	ASD↓
Full PMRNet	**91.02**	**95.23**	**13.99**	**4.08**
w/o LMB	90.61	94.97	15.61	4.46
w/o PGRF	90.43	94.89	14.84	4.38
w/o BF	90.24	94.78	15.50	4.43
w/o All Three	89.80	94.52	16.98	4.69

As shown in Table 3, removing BF suffers the worst boundary degradation among single-module removals, which shows that the decoder relies on explicit boundary features. When we remove LMB, the IoU drop is small (0.41%), but HD95 jumps by 1.62. This gap implies that prototype matching [56] contributes less to region overlap and more to keeping boundary localization consistent across scales in this case. Without PGRF, all four metrics degrade evenly, consistent with a global context prior [47] that benefits the prediction as a whole rather than single aspect. Removing all three together drops IoU to 89.80% and pushes HD95 to 16.98, pointing that the three modules are not merely independently useful but mutually reinforcing.

5.2. PGRF Replacements

In Table 4, we compare PGRF with three context aggregation alternatives on the Clinic [4] dataset: Large Kernel [16] (two stacked 7×7 depthwise separable convolutions), Dilated

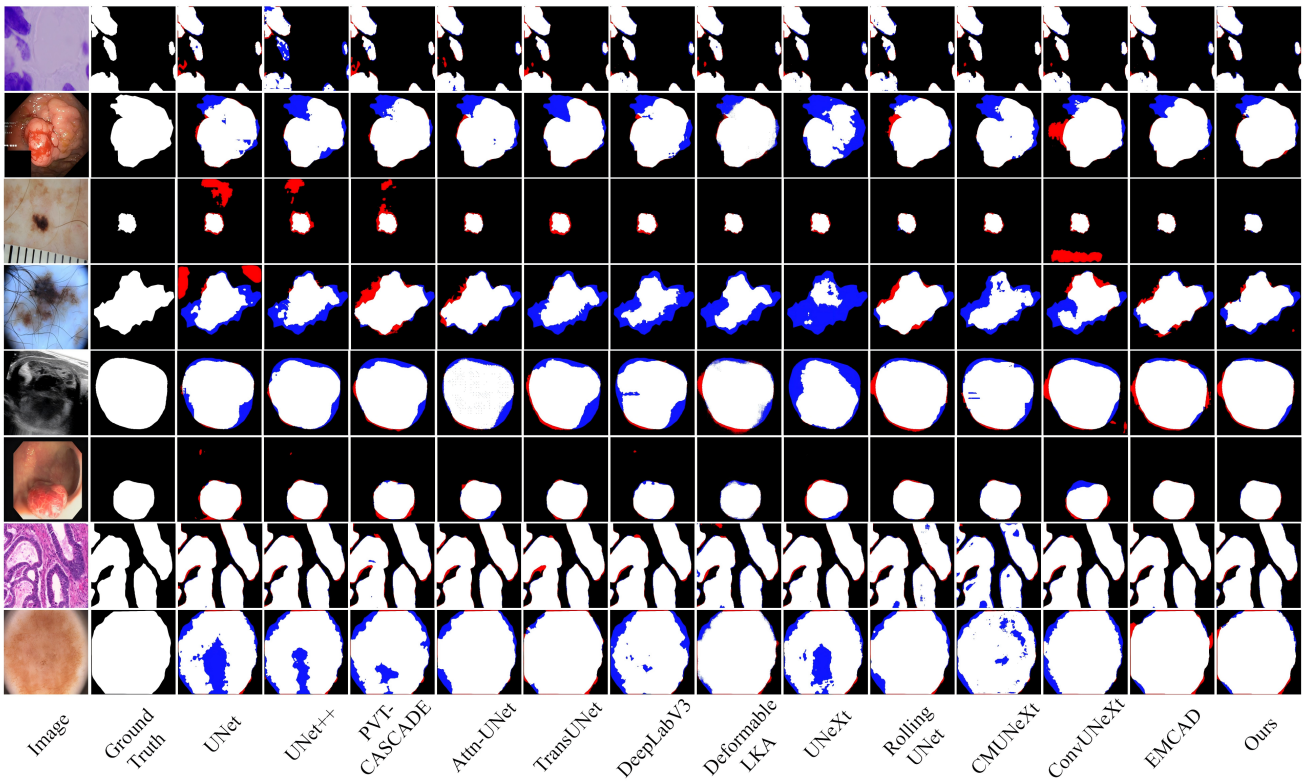


Figure 4. Qualitative comparison of segmentation results across different methods on various datasets. Our PMRNet produces more accurate boundary delineation and handles small objects better than baseline methods, particularly evident in challenging cases with low contrast and ambiguous boundaries. Red: false positive regions; Blue: false negative regions.

Conv [68] (parallel 3×3 convolutions with dilation rates 2, 4, 6, fused by a 1×1 convolution), and Global Pooling (global average pooling followed by an MLP). PGRF leads all variants, with IoU gains of 1.11%, 1.43%, and 1.68% over the three alternatives respectively, and consistent improvements in HD95 ranging from 0.39 to 1.23.

Table 4. Replacing PGRF with context aggregation alternatives on Clinic [4] dataset.

Variant	IoU(%)↑	Dice(%)↑	HD95↓	ASD↓
Large Kernel (7×7 DSConv $\times2$) [16]	86.14	91.82	13.52	3.68
Dilated Conv (3×3, d=2,4,6) [68]	85.82	91.50	13.87	3.79
Global Pooling (GAP+MLP)	85.57	91.18	14.36	3.95
PGRF (Ours)	**87.25**	**92.56**	**13.13**	**3.54**

5.3. ESConv Replacements

Table 5 reports results of replacing ESConv with Edge Attention [5], Sobel Filter [36] + Conv, and Canny Edge [7] + Conv on ISIC2017 [14]. ESConv achieves 83.14% IoU and 13.73 HD95, outperforming the closest competitor (Edge Attention) by 0.93% IoU and 1.35 HD95. The gap widens further against fixed-operator methods, reaching 1.51% IoU for Canny Edge + Conv. The margin over both learned and fixed-operator baselines suggests that the symplectic coupling provides a more effective boundary prior than either alternative alone.

Table 5. Replacing ESConv with boundary enhancement alternatives on ISIC2017 [14] dataset.

Variant	IoU(%)↑	Dice(%)↑	HD95↓	ASD↓
Edge Attention [5]	82.21	88.93	15.08	5.93
Sobel Filter [36] + Conv	81.87	88.71	15.62	6.09
Canny Edge [7] + Conv	81.63	88.54	15.91	6.18
ESConv (Ours)	**83.14**	**89.69**	**13.73**	**5.54**

6. Conclusion

In this paper, we have presented PMRNet, a lightweight encoder-decoder network for medical image segmentation. Instead of relying on scaling up model capacity, PMRNet adopts a new strategy [2, 47, 65] to address common limitations in existing segmentation architectures. With only 0.87M params, 3.43 GFLOPs without pre-training, PMRNet maintains high inference speed and strong segmentation accuracy across 12 datasets covering diverse clinical scenarios, enhancing its practicality for real-world application. Future work will explore PMRNet in volumetric (3D) segmentation and investigate its integration with foundation models [43].

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